

BROOKINGS, SD, USA

WMO: 726515

Lat: 44.300N

Lon: 96.800W

Elev: 502

StdP: 95.43

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.7	-32.2	0.2	-22.2	-28.8	0.3	-21.4	14.3	-10.6	12.6	-10.3	3.4	320	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.0	23.2	30.0	22.1	28.2	21.0	24.4	29.8	23.4	28.4	22.3	26.9	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	18.4	27.8	22.0	17.7	27.2	20.9	16.6	26.2	76.5	30.1	72.3	28.1	68.0	27.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.8	9.3	-29.9	34.6	3.1	1.5	-32.1	35.7	-33.9	36.6	-35.7	37.4	-37.9	38.5		
(5)			-29.2	26.0	2.8	1.1	-31.3	26.8	-32.9	27.4	-34.5	28.0	-36.5	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.9	-7.3	-0.9	6.6	13.5	19.2	22.0	20.2	15.9	8.3	0.4	-7.0
	DBStd	12.39	7.22	7.31	7.18	5.65	4.84	3.84	3.30	3.21	4.64	5.32	6.00	6.64
	HDD10.0	2517	618	486	342	129	21	0	0	7	95	292	527	
	HDD18.3	4503	877	719	596	352	162	35	8	18	100	312	539	786
	CDD10.0	1359	0	0	5	28	131	276	372	317	184	43	3	0
	CDD18.3	302	0	0	0	1	13	60	122	76	27	2	0	0
	CDH23.3	2746	0	0	2	22	160	556	1125	609	244	29	0	0
	CDH26.7	797	0	0	0	4	44	160	371	158	54	4	0	0
(14) Wind	WSAvg	4.4	4.4	4.6	4.9	5.3	4.8	4.1	3.8	3.7	4.2	4.6	4.5	4.4
(15) Precipitation	PrecAvg	578	8	11	31	54	72	107	82	74	66	41	21	9
	PrecMax	820	30	60	88	136	236	236	307	177	195	163	113	53
	PrecMin	337	0	0	6	7	11	21	20	24	2	0	0	0
	PrecStd	128	9	12	23	30	50	56	60	37	44	38	23	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.9	21.0	26.4	31.1	32.8	34.1	32.5	31.0	27.2	19.1	10.0
		MCWB	3.6	8.3	13.3	15.6	19.4	22.8	24.1	23.7	21.5	17.0	11.6	5.2
	2%	DB	3.0	7.6	16.2	22.3	27.4	30.3	32.3	30.2	28.0	22.6	15.8	6.3
		MCWB	0.9	4.1	11.0	13.0	17.6	21.5	23.9	22.9	20.1	15.3	9.7	2.8
	5%	DB	1.8	4.4	12.6	18.8	24.6	28.3	30.9	28.0	26.3	19.1	12.2	3.0
		MCWB	0.5	2.0	8.6	11.2	16.2	20.6	23.2	21.4	18.9	13.0	7.5	0.8
	10%	DB	0.0	2.5	8.9	16.2	22.3	27.0	28.7	27.1	24.0	17.2	8.8	1.7
		MCWB	-1.0	0.5	5.9	9.5	14.9	19.4	22.0	20.6	18.0	12.1	5.1	0.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	8.7	14.6	16.4	20.9	24.6	25.9	25.2	23.3	19.1	13.1	5.9
		MCDB	7.3	13.4	18.3	24.3	28.5	30.3	31.3	30.4	27.7	24.3	17.7	9.1
	2%	WB	1.4	4.2	11.6	14.0	18.9	23.0	24.8	23.9	21.7	16.4	10.1	3.0
		MCDB	3.0	7.1	15.2	20.7	25.5	28.5	30.6	28.6	26.2	20.9	14.6	5.8
	5%	WB	0.3	2.4	8.8	12.0	17.4	21.5	23.8	22.8	20.5	14.4	7.8	1.2
		MCDB	1.6	4.8	12.7	17.2	23.0	27.2	29.1	27.1	25.1	18.9	11.9	3.1
	10%	WB	-1.1	0.5	5.7	10.4	15.7	20.1	22.8	21.6	19.1	12.3	5.4	0.0
		MCDB	-0.1	2.3	8.8	15.4	20.8	25.8	27.6	25.6	23.1	16.3	8.6	1.5
(35) Mean Daily Temperature Range	MDBR		9.2	9.7	9.5	11.1	11.2	10.8	11.1	10.9	11.6	11.0	10.0	8.8
	5% DB	MCDBR	10.7	12.0	14.5	16.5	15.1	12.9	12.5	12.2	13.9	15.7	15.3	11.3
		MCWBR	8.8	9.2	9.8	9.3	8.2	6.8	6.5	6.9	7.5	9.5	10.1	8.7
	5% WB	MCDBR	10.2	11.6	14.0	14.3	13.1	11.7	11.2	11.0	11.6	13.3	14.6	10.8
		MCWBR	8.5	8.9	9.6	8.9	7.7	6.9	6.5	6.8	7.0	8.7	10.0	8.5
(40) Clear-Sky Solar Irradiance	taub		0.263	0.286	0.315	0.364	0.378	0.388	0.395	0.398	0.352	0.321	0.289	0.267
	taud		2.437	2.394	2.404	2.298	2.339	2.367	2.343	2.315	2.420	2.483	2.482	2.442
	Ebn at Noon		876	910	924	898	892	881	870	854	871	858	835	832
	Edh at Noon		71	89	102	123	122	120	121	120	100	82	68	64
(44) All-Sky Solar Radiation	RadAvg		1.86	2.91	3.79	4.81	5.59	6.23	6.68	5.51	4.34	2.88	1.94	1.43
	RadStd		0.19	0.28	0.37	0.46	0.42	0.43	0.24	0.45	0.33	0.35	0.17	0.12

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page